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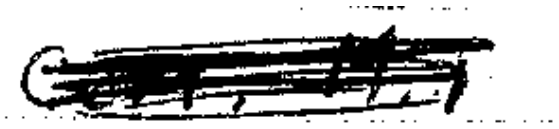
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Chapter 13

GELFAND-P000777

27/III–78. [Crossed out: C...]



Undeciphered passage. An initial C remains between the date and the heading; dense cancellation strokes obscure the rest.

Original source details: GELFAND-P000777

Gelfand Seminar.

F. Calogero. On the Bäcklund transform.

(all of this jointly with A. Degasperis)

$$\Psi_{xx}(x,k) = [q(x)\ [?]\ k^2]\Psi(x,k), \qquad -\infty < x < +\infty \tag{1}$$



Uncertain reading. Heavy overwriting obscures the sign between $q(x)$ and k^2; its reading is unresolved.

Original source details: GELFAND-P000777

$$q(x) \to 0 \quad \text{as } |x| \to \infty \quad \text{sufficiently rapidly} \tag{2}$$

Let Ψ — a solution such that

$$\begin{cases} \Psi(x,k) \underset{x\to+\infty}{\longrightarrow} e^{-ikx} + R(k)e^{ikx}, \\ \Psi(x,k) \underset{x\to-\infty}{\longrightarrow} T(k)e^{-ikx} \end{cases} \tag{3}$$

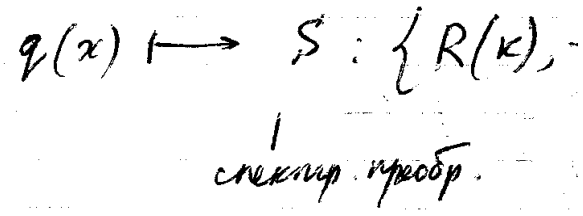


Diagram and typed description.

$q(x) \longmapsto S \quad : \{R(k),\ -\infty < k < +\infty;$

spectral transform

and some discrete spectrum information$\}$.

Original source details: GELFAND-P000777

For a wide class of potentials, the discrete spectrum = the poles of $R(k)$:

$$R(k) \sim \frac{ic_n^2}{k - i\rho_n} \tag{4}$$

The potential q is determined by $R(k)$.

[Crossed out:]

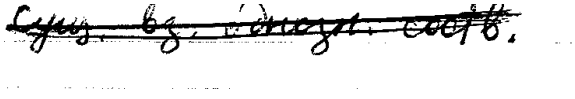


Diagram and typed description.

~~There exists a one-to-one correspondence.~~

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Let [crossed out: $q(x)$]



Uncertain reading. The heavily cancelled expression tentatively reads $q(x)$.

Original source details: GELFAND-P000777

$q = q(x,t)$, $R = R(k,t)$.

In which cases

R changes simply, while q changes interestingly?

[At the left edge of the frame, beyond the margin line: ...a))]



Uncertain reading. The image edge clips the entry; the end of an a-like letter and two curved strokes remain visible.

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Technique: since $R(k)$ is determined by asymptotics, it is natural to use Wronskians.

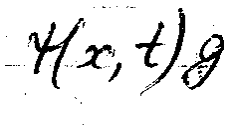
2 equations. Let there be $q, q' \longleftrightarrow R, R'$.

$$2ikf(-4k^2)[R'(k) - R(k)]$$

$$= \int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,k) f(\Lambda)[q'(x) - q(x)] \quad (5)$$

$$(2ik)^2 g(-4k^2)[R'(k) + R(k)]$$

$$= \int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,t) g(\Lambda)\Gamma \cdot 1 \quad (6)$$

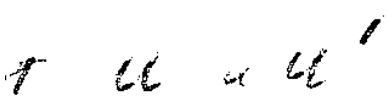


Uncertain reading. The second factor in (6) visibly reads $\Psi(x,t)$, although the first has argument k.

Original source details: GELFAND-P000777

GELFAND-P000778

Λ — an integro-differential operator depending on u and u'



Uncertain reading. The opening definition has u and u', unlike the q, q' in (7), (8).

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$\underline{\Gamma}$ — also an operator

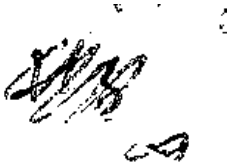
Here

$$\underline{\Lambda}F(x) = F_{xx}(x) - 2[q'(x) + q(x)]F(x) + \underline{\Gamma}\int_x^\infty dx'\, F(x'); \tag{7}$$

$$\underline{\Gamma}G(x) = [q'_x(x) + q_x(x)]G(x) + [q'(x) - q(x)]\int_x^\infty dx'[q'(x') - q(x')]G(x'); \tag{8}$$

f, g — arbitrary functions.

[Crossed out: ...]



Uncertain reading. Slanting strokes cancel the short entry before “Property”.

Original source details: GELFAND-P000778

Property of $\underline{\Lambda}$:

$$\int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,k)\underline{\Lambda}F(x) = -4k^2\int_{-\infty}^{+\infty} dx\, \Psi'(x,k)\Psi(x,k)F(x). \tag{9}$$

Let q and q' be distinct (a limit will be taken later).

If q and q' are such that

$$f(\underline{\Lambda})[q'(x) - q(x)] + g(\underline{\Lambda})\underline{\Gamma}\,1 = 0, \tag{10}$$

then

$$R'(k) = \frac{f(-4k^2) - 2ikg(-4k^2)}{f(-4k^2) + 2ikg(-4k^2)}R(k) \tag{11}$$

[Crossed out: ...]



Uncertain reading. The short entry to the left of the parenthesis is cancelled; its reading is unresolved.

Original source details: GELFAND-P000778

(this follows from formulas (5), (6))

All of this is for the stationary Schrödinger equation.

Now one must take infinitesimal deformations

$$q = q(x,t), \qquad q' = q(x, t+dt)$$

$$\begin{aligned} R' - R &\longmapsto R_t(k,t) \\ q' - q &\longmapsto q_t(x,t) \\ \underline{\Lambda} &\longmapsto \underline{L} \qquad (= \underline{\Lambda}|_{q'=q}) \end{aligned}$$

$\underline{\Gamma}$ [crossed out: and?]



Uncertain reading. The cancelled entry between the operator sign and the arrow tentatively reads “и” (and).

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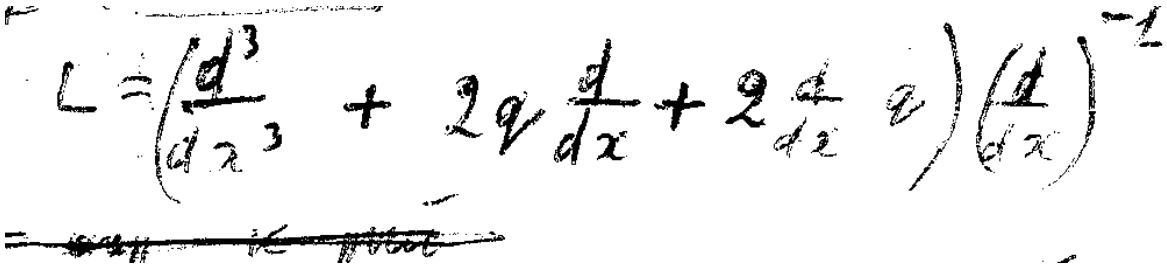
$\longmapsto$ multiplication by $2q_x(x)$

We obtain:

$$\begin{aligned} \underline{L}F(x) = F_{xx}(x) - 4q(x,t)F(x) \\ + 2q_x(x,t)\int_x^\infty dx'\, F(x') \end{aligned} \tag{12}$$

Gelf. (and Dickey):

$$L = \left(\frac{d^3}{dx^3} + 2q\frac{d}{dx} + 2\frac{d}{dx}q\right)\left(\frac{d}{dx}\right)^{-1}$$



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Uncertain reading. In this entry both signs before the terms containing q are “+”.

[Crossed out:]

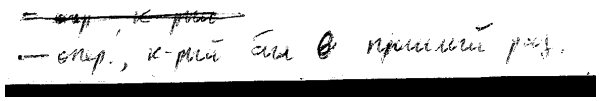


Diagram and typed description. ~~= the operator which~~

Original source details: GELFAND-P000778

— the operator which was present last time.

GELFAND-P000779

[source number: 5]

appears in many places.

a) [Crossed out: and?]

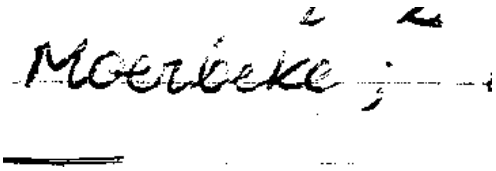


Uncertain reading. The short cancelled sign after a) tentatively reads “и” (and).

Original source details: GELFAND-P000779

Ψ_1, Ψ_2 satisfy a second-order equation $\Rightarrow$ $\Psi_1\Psi_2$ satisfies a third-order equation.

Gelf. Dickey; Lax; Marchenko; Mose...ka; et al.



Uncertain reading. The surname after Marchenko starts with “Moce...” and ends with ka-like letters; the complete reading is unresolved.

Original source details: GELFAND-P000779

Property of $\underline{\Lambda}$.

$$\int_{-\infty}^{\infty} dx\, \Psi'(x,k)\Psi(x,k)(\underline{\Lambda} + 4k^2)F(x) = 0 \tag{13}$$

$\Psi = \Psi' \Rightarrow \underline{\Lambda}$ is replaced by $\underline{L}$.

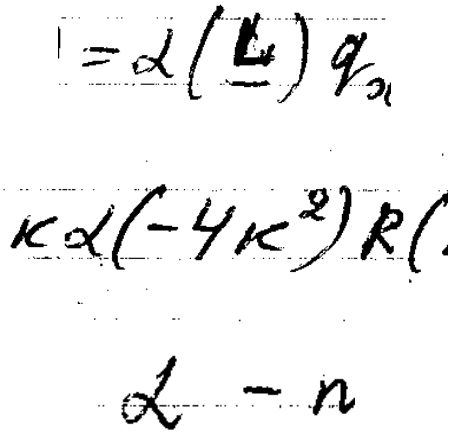
[Above the following line: a row of dots.]

. . . . . .

For $f(\underline{L}) = \underline{L}$ we have

$$\begin{gathered} q_t(x,t) = \alpha(\underline{L})q_x(x,t) \\ \Longleftrightarrow R_t(k,t) = 2ik\alpha(-4k^2)R(k,t) \end{gathered} \tag{14}$$

α — a polynomial $\Rightarrow$ KdV and higher KdV



Uncertain reading. The sign is transcribed as α in all three places; its shape also permits a calligraphic L.

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$R' \leftrightarrow R$ by formula (11) and R — a solution of the linear equation $(14)_2$, then R' is also a solution.

Then q', constructed from q, satisfies equation $(14)_1$ together with q.

$(14)_1$ — higher KdV in the form $AKNS$

The simplest case: let [crossed out: $f, g =$ const] $g = 1$, $f = 2\rho$.



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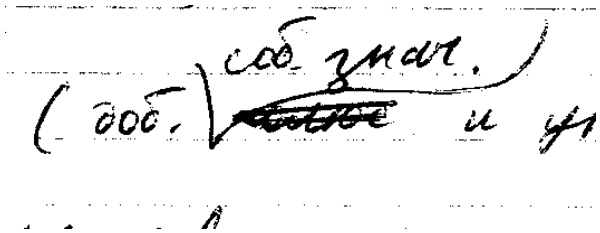
Diagram and typed description.

~~$f, g =$ const~~

One obtains the simplest transformation:

$$R'(k,t) = -\frac{k+i\rho}{k-i\rho} R(k,t) \tag{15}$$

This adds a soliton (adds [crossed out: ...; inserted above with a connecting line: eigenvalue] and multiplies by a coefficient)



Original source details: GELFAND-P000779

Diagram and typed description.

eigenvalue

...

Heavy cancellation strokes obscure the word beneath the insertion; its reading is unresolved.

The relation between the potentials is written out:

let

$$Q(x,t) = \int_x^\infty dx'\, q(x',t) \tag{16}$$

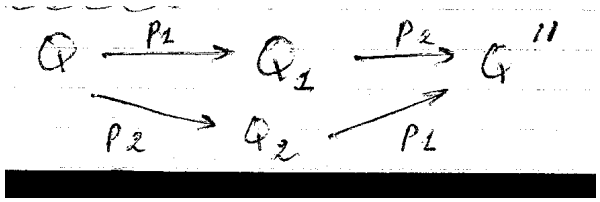
$$\begin{aligned} Q'_x(x,t) + Q_x(x,t) = & -\tfrac{1}{2}[Q'(x,t) - Q(x,t)] \\ & \cdot [4\rho + Q'(x,t) - Q(x,t)] \end{aligned} \tag{17}$$

$Q \equiv 0 \Rightarrow Q'$ — a one-soliton solution, (for $q_t = q_{xxx} - 6q_x q$)

(17) — a Riccati equation (with respect to Q' or with respect to Q)

All integrals are obtained by successive application of the Bäcklund transform from the trivial $\int u^2\, dx$

Another application



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Diagram and typed description.

$$\begin{array}{ccccc} Q & \xrightarrow{\rho_1} & Q_1 & \xrightarrow{\rho_2} & Q'' \\ & \searrow^{\rho_2} & & \nearrow_{\rho_1} & \\ & & Q_2 & & \end{array} \tag{18}$$

commutativity!

This is immediately visible; for the functions R it is simply multiplication. For the functions q it is nontrivial. One obtains a principle of nonlinear superposition:

$$Q''(x,t) = Q(x,t) - \frac{(\rho_1+\rho_2)[Q_1(x,t)-Q_2(x,t)]}{\{\rho_1-\rho_2+\frac{1}{2}[Q_1(x,t)\ [?]\ Q_2(x,t)]\}} \tag{19}$$

[In the manuscript the fraction in (19) is written with a slash at the end of the first line and the start of the second.]



Uncertain reading. Heavy overwriting obscures the sign between $Q_1(x,t)$ and $Q_2(x,t)$; $+$ and $-$ are possible.

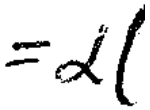
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— · —

f and g may depend on t! The relation between (10) and (11) remains valid.

$$q_t(x,t) = \alpha(\underline{L})q_x(x,t) \tag{20}$$

$$R(k,t) = e^{2ik\alpha(-4k^2)(t-t_0)}R(k,t_0) \tag{21}$$



Uncertain reading. The sign is provisionally rendered α in (20), (21); the handwriting also permits $\mathcal{L}$.



Original source details: GELFAND-P000780

Choose f and g so that the fraction in (11) coincides with the displayed exponential in (21). Then we obtain a correspondence between $q(x,t)$ and $q(x,t_0)$ (formula (10)).

Evolution in t — also a Bäcklund transform!

There is an analogy with the Fourier transform:

$$u(x,t) = e^{(t-t_0)\psi(\frac{\partial}{\partial x})}u(x,t_0),$$

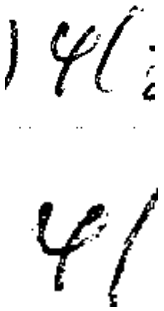


Uncertain reading. The first u overwrites an earlier unreadable sign.

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when

$$u_t(x,t) = \psi\left(\frac{\partial}{\partial x}\right) u(x,t).$$



Uncertain reading. Both lines provisionally have ψ; another possible reading is φ.

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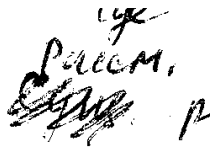
— · —

Consider the equation

$$\Psi_{tt} + \tfrac{1}{2}\Psi_{xx} - x\Psi_x + n\Psi = 0, \tag{22}$$

where $n \in \mathbb{Z}$

[Crossed out, tentatively: Its. Inserted above: Consider.]



Uncertain reading. The cancelled word may read “Его” or “Её” (its).

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a solution (of the form):

$$\Psi(x,t) = \prod_{j=1}^{n}(x - x_j(t)) \tag{23}$$

[Crossed out to the right of (23):]

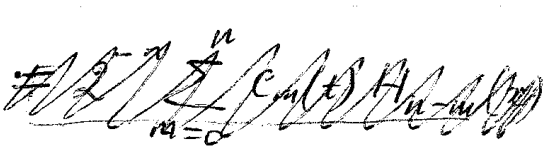


Diagram and typed description.

$$\cancel{= 2^{-n} \sum_{m=0}^{n} c_m(t) H_{n-m}(x)}$$

Original source details: GELFAND-P000780

Then

$$\ddot{x}_j = -x_j(t) + \sum_{k=1}^{n}{}' \frac{1 + 2\dot{x}_j(t)\dot{x}_k(t)}{x_j(t) - x_k(t)} \tag{24}$$

(Moser) [A word is crossed out at right; only a possible initial M or H is distinguishable.]



Undeciphered passage. The crossed-out word to the right of “Мозер” (Moser) cannot be read.

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[Inserted beneath, one below the other, with a brace at right:]

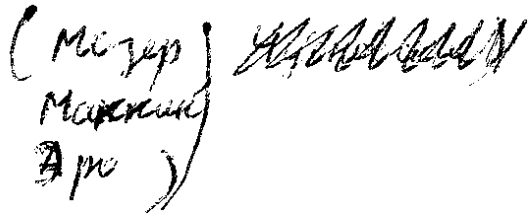


Diagram and typed description.

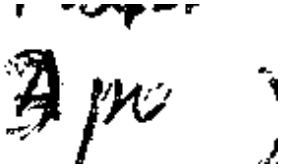
(Moser)
Makkin
Ero

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Uncertain reading. “Маккин” (Makkin) is tentative.

Original source details: GELFAND-P000780



Uncertain reading. “Эро” (Ero) is tentative.

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GELFAND-P000781

[source number: 6]

Another form:

$$\Psi(x,t) = 2^{-n} \sum_{m=0}^{n} c_m(t) H_{n-m}(x) \tag{25}$$

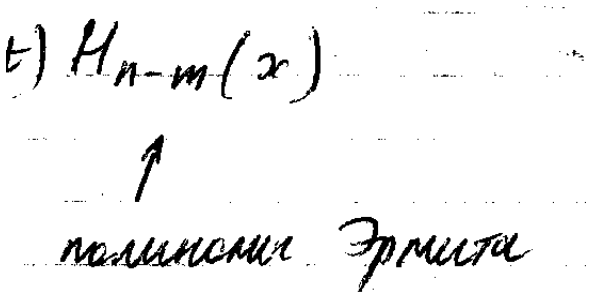


Diagram and typed description.

$H_{n-m}(x)$
↑
Hermite polynomials

Original source details: GELFAND-P000781

Then

$$\ddot{c}_m(t) + m c_m(t) = 0 \tag{26}$$

Nonlinear correspondence:

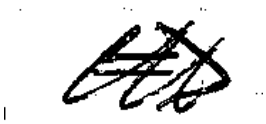
$$\{x_j\} \leftrightarrow \{c_m\}$$

From roots to coefficients — one obtains a linear system of differential equations.

Equilibrium solutions of (24)

$$\bar{x}_j = \sum_{k=1}^{n}{}'(\bar{x}_j - \bar{x}_k)^{-1} \tag{27}$$

In this case $c_m \equiv 0$, [Crossed out: ...]



Uncertain reading. The short mathematical entry is heavily cancelled; it may read $\forall t$.

Original source details: GELFAND-P000781

$m = 1, 2, \ldots, n$; $c_0 = 1$

the equilibrium positions are the zeros of the Hermite polynomial of order n

Corollary:

the zeros of the Hermite polynomial satisfy (27).

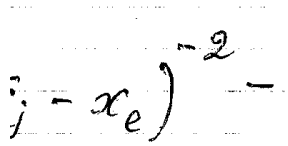
(Stieltjes; 1885)

Without Ψ_{tt} equation (22) — the equation for Hermite polynomials.

Small oscillations about the equilibrium positions:

$$x_j = \bar{x}_j + \varepsilon_j(t) \tag{28}$$

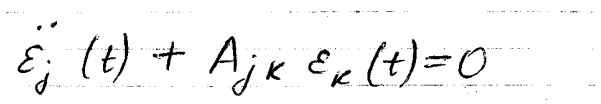
$$A_{jk} = \delta_{jk} \sum_{\ell=1}^{n}{}'(\bar{x}_j - x_\ell)^{-2} - (1 - \delta_{jk})(\bar{x}_j - \bar{x}_k)^{-2}$$



Uncertain reading. The first term has x_ℓ without a visible bar; the other term has visible bars.

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$$\ddot{\varepsilon}_j(t) + A_{jk}\varepsilon_k(t) = 0$$



Uncertain reading. The equation has $A_{jk}\varepsilon_k(t)$, with no separate term $\varepsilon_j(t)$.

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The eigenvalues are integers —

$$Av^{(s)} = a_s v^{(s)}, \qquad a_s = s - 1, \quad s = 1, 2, \ldots, n.$$

— · —

GELFAND-P000782

3/IV-78 Gelfand Sem.

D. B. Fuks (and I. M. Gelfand)

Computation of the cohomology of various Lie algebras.

g — a real Lie algebra;

$C_q(g)$ — the group of q-dimensional chains

$$C_q(g) \underset{\text{def}}{=} \Lambda^q g \qquad \text{(exterior power)}$$

$$d : C_q(g) \longrightarrow C_{q-1}(g) \tag{1}$$

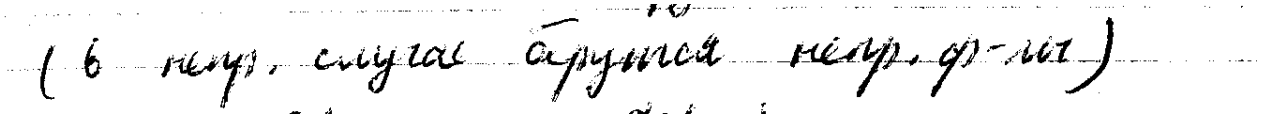
$$d(g_1 \wedge \ldots \wedge g_q) = \sum_{1 \le s < t \le q} (-1)^{s+t-1} [g_s, g_t] \wedge g_1 \wedge \ldots \wedge \hat{g}_s \wedge \ldots \wedge \hat{g}_t \wedge \ldots \wedge g_q \tag{1'}$$

GELFAND-P000783

[source number: 11]

It is easy to see that $d^2 = 0$ and this gives rise to homology, denoted by $H_q(g)$.

The dual construction gives cohomology (in the continuous case one takes continuous



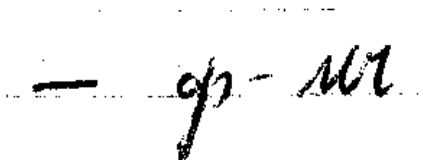
Original source details: GELFAND-P000783

Uncertain reading. Abbreviation: “ф-лы” / “ф-ии” (functionals / functions).

)

$$\delta : C^{q-1}(g) \longrightarrow C^q(g) \tag{2}$$

$C^q(g)$ —



Original source details: GELFAND-P000783

Uncertain reading. Abbreviation: “ф-лы” / “ф-ли” (functionals).

on $C_q(g)$

$H^q(g)$ — the homology of the dual complex with differential δ;

An inner product lets us identify C^q and C_q. We obtain:

$$d : C_q(g) \longrightarrow C_{q-1}(g)$$
$$\delta : C_q(g) \longrightarrow C_{q+1}(g)$$

Put

$$\Delta = d\delta + \delta d \tag{3}$$

(Laplace operator)

Δ commutes with d and δ.

It is easy to show that each homology class has exactly one harmonic element.

$$\Delta a = 0 \quad \Longleftrightarrow \quad da = 0 \text{ and } \delta a = 0 \tag{4}$$

Consider the Lie algebra [inserted above the line: (formal)] of vector fields on $\mathbb{R}$ of the form:

$$f(x)\frac{d}{dx}, \tag{5}$$

where

$$f(x) = \sum_{k\geq 2} a_k x^k. \tag{6}$$

Put

$$e_n = x^{n+1}\frac{d}{dx} \qquad \text{(element of "dimension" } n\text{)}$$

The cohomology was computed by L. Goncharova, but since then there have been further attempts to understand this.

The algebra W_1 does not even have a local Lie group

[Below W_1 is written $\{a(x)\frac{d}{dx}\}$; two short vertical strokes link the inscription to W_1.]

[Crossed out: ...]



Uncertain reading. A short cancelled word; possibly "Но" or "Ново".

Original source details: GELFAND-P000783

The group of formal diffeomorphisms has Lie algebra: $\{a(x)\frac{d}{dx},\ a(0)=0\}$

$$L_0 \underset{\text{def}}{=} \left\{a(x)\frac{d}{dx},\ a(0)=0\right\} \tag{7}$$

— it is the Lie algebra of the formal group

$$W_1 \underset{\text{def}}{=} \left\{a(x)\frac{d}{dx},\ a \text{ — a formal series}\right\} \tag{8}$$

$$L_1 \underset{\text{def}}{=} \left\{a(x)\frac{d}{dx},\ a(0)=0,\ a'(0)=0\right\} \tag{9}$$

[At the lower left, beyond the edge of the sheet:



Uncertain reading. a cropped fragment of a sign

Original source details: GELFAND-P000783

.]

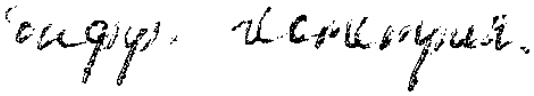
GELFAND-P000784

$$L_0 \subset W_1, \qquad \dim W_1/L_0 = 1.$$

L_0 integrates to a group

W_1 does not integrate to a group.

This is why there is this whole



Uncertain reading. “следу.” (abbreviation, probably “following”).

Original source details: GELFAND-P000784

story.



Uncertain reading. “Спрв.” / “Справ.” (abbreviated heading) 1.

Original source details: GELFAND-P000784

A Lie alg. is called an algebra of finite defect, if it has a subalgebra of finite codimension integrating to a Lie group. The minimum codimension is called the defect.

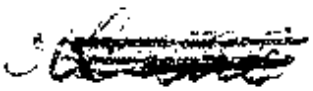
How it began: automorphism groups.



Uncertain reading. “Спрв.” / “Справ.” (abbreviated heading) 2.

Original source details: GELFAND-P000784

Suppose there is no centre; then the automorphism group [crossed out:



Uncertain reading. A short word, possibly “имеет” (“has”).

Original source details: GELFAND-P000784

] has a Lie algebra that is an



Uncertain reading. overalgebra / subalgebra

Original source details: GELFAND-P000784

of the algebra itself.

The cohomology of W_1 and L_0 is easily found (Gelfand, Fuks); but that of L_1 is harder.

[A short underlined inscription with diagonal hatching:



Uncertain reading. “Фукс;” / “Факт;” (Fuks; / Fact;)

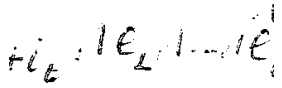
Original source details: GELFAND-P000784

.]

L_1 is generated by the elements $e_1, e_2, \ldots$

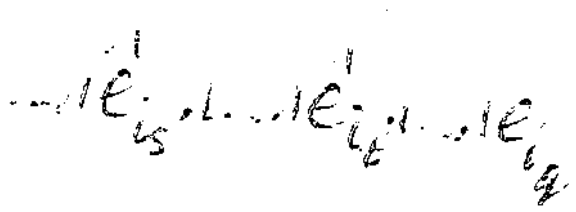
$$[e_n, e_m] = (m-n)e_{m+n} \tag{10}$$

$$d(e_{i_1} \wedge \cdots \wedge e_{i_q}) = \sum_{s<t} (-1)^{s+t-1}(i_t - i_s)e_{i_s+i_t} \wedge e_1 \wedge \cdots \wedge \overset{\prime}{e}_{i_s} \wedge \cdots \wedge \overset{\prime}{e}_{i_t} \wedge \cdots \wedge e_{i_q} \tag{11}$$



Uncertain reading. After $e_{i_s+i_t}$ the source has e_1, with no visible i in the subscript.

Original source details: GELFAND-P000784



Uncertain reading. Above e_{i_s} and e_{i_t} there are short almost vertical strokes; the shape of the marks is not entirely clear.

Original source details: GELFAND-P000784

We see that the sum of the indices does not change! We obtain a



Uncertain reading. “расслоение” / “разложение” (splitting / decomposition)

Original source details: GELFAND-P000784

into a direct sum of finite-dimensional complexes.

Theorem 1 (L. Goncharova)

a) $\forall q \geq 1 \quad \Rightarrow \quad \dim H_q(L_1) = 2$

b) «weight» of the generators

$$\frac{3q^2 \pm q}{2}$$

(sum of the indices)

— · —

$C_{q,n}$ — q-chains of weight n. There is a Laplace operator:

$$\Delta : C_{q,n} \longrightarrow C_{q,n} \tag{12}$$

[Introduce another structure]



Uncertain reading. The line has diagonal hatching; whether this is emphasis or cancellation is unclear.

Original source details: GELFAND-P000784

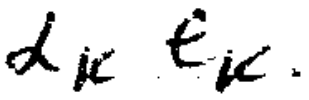
δ is the adjoint with respect to the inner product in which the monomials $e_{i_1} \wedge \cdots \wedge e_{i_q}$ form an orthonormal basis.

GELFAND-P000785

[source number: 12]

3 candidates for an orthonormal basis:

$$e_k, \qquad \frac{e_k}{k!}, \qquad \alpha_k e_k.$$



Uncertain reading. The coefficient in the third candidate: α_k / λ_k.

Original source details: GELFAND-P000785

Here we use e_k

This carries over to the exterior powers.

We obtain the Laplace operator (12).

Facts



Original source details: GELFAND-P000785

Uncertain reading. After the heading there is a tall hatched sign, possibly III.

A. All] eigenvalues — integers!



Original source details: GELFAND-P000785

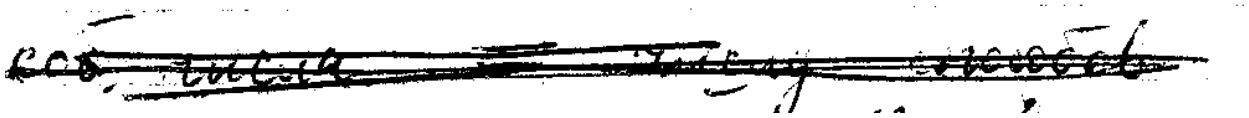
Uncertain reading. After “All” there is a tall sign resembling a closing square bracket.

B. they can be enumerated!

In the space

$$C_{*,n} = \bigoplus_q C_{q,n}$$

[Crossed out: eigenvalues ...



Original source details: GELFAND-P000785

Undeciphered passage. The middle of the phrase is illegible; the right end may read “способ...”.

]

The number of distinct eigenvalues $=$
$=$ the number of partitions of n into a sum

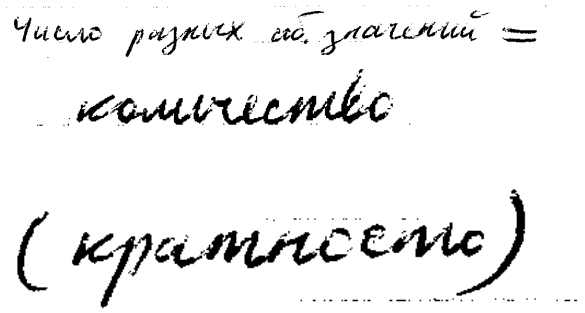
$$n = i_1 + \cdots + i_q \tag{13}$$

$$\begin{cases} i_1 \geq 1 \\ i_2 - i_1 \geq 3 \\ \cdots \\ i_q - i_{q-1} \geq 3 \end{cases} \tag{14}$$

To the partition (13) there corresponds the quantity

$$C_{i_1}^3 + \cdots + C_{i_q}^3 - \sum_{s<t} i_s i_t \tag{15}$$

of eigenvalues (multiplicity)



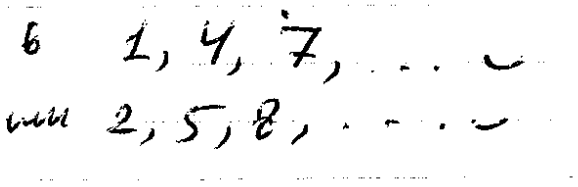
Uncertain reading. The words “quantity” and “of eigenvalues (multiplicity)” translate the literal source wording.

Original source details: GELFAND-P000785

Exercise:

the number (15) ≥ 0 and $= 0$

$$\Longleftrightarrow \quad \begin{array}{lll} \text{in } 1,4,7,\ldots & \smile & \Sigma = \dfrac{3q^2 - q}{2} \\ \text{or } 2,5,8,\ldots & \smile & \Sigma = \dfrac{3q^2 + q}{2} \end{array}$$



Uncertain reading. After each sequence there is a short upward-opening arc; the meaning of the marks is unclear.

Original source details: GELFAND-P000785

The eigenvectors can be written down

Symbolic calculus:

$$z = e_1, \quad e_2 = z^2, \quad e_3 = z^3; \ \ldots$$

Description of an eigenvector:

$$\left(\prod_{1 \leq i < j \leq q} (z_i - z_j) \right)^3$$

multiply by

$$\prod z_i \qquad \text{or} \qquad \prod z_i^2$$

One obtains an antisymmetric polynomial. It is represented as a combination

$$z_1^{i_1} \cdots z_q^{i_q} + \underbrace{\cdots}_{\text{with the required signs}}$$

Replace by

$$e_{i_1} \wedge \cdots \wedge e_{i_q}$$

Next meeting 4/IV-78